Chapter 4: Expressions Using Letter-Numbers

Worksheet 4: Expressions Using Letter-Numbers

Learning Objective: Understand variables as letter-numbers, construct and evaluate algebraic expressions, and identify terms, coefficients and constants.

Worked Example:
If a pencil costs Rs. x, write an expression for the cost of 5 pencils, and find the cost when $x = 8$.

Solution: Expression $= 5x$. When $x = 8$, cost $= 5 \times 8 =$ Rs. 40.

Questions (1–20)

1. In the expression $7x + 3$, what is the coefficient of x?

2. In the expression $4y - 9$, what is the constant term?

3. Write an expression for "5 more than a number n".

4. Evaluate $3x + 7$ when $x = 4$.

5. Write an expression for "three times a number m, decreased by 6".

6. A rectangle has length l and breadth b. Write an expression for its perimeter.

7. A notebook costs Rs. p and a pen costs Rs. q. Write an expression for the cost of 3 notebooks and 2 pens.

8. Evaluate $5a - 2b$ when $a = 6$ and $b = 3$.

9. Identify the terms in the expression $4x + 3y - 7$.

10. Write an expression for "the sum of twice a number x and 5".

11. A father's age is 4 times his son's age. If the son's age is s years, write an expression for the father's age, and find it when $s = 9$.

12. Simplify: $5x + 3x - 2x$.

13. Simplify: $7a + 4b - 2a + b$.

14. A shopkeeper sells notebooks at Rs. n each. Write an expression for the earnings from selling 25 notebooks, and find the earnings if $n = 15$.

15. Evaluate $2(x+3)$ when $x=5$.

16. A taxi ride costs Rs. 30 plus Rs. 8 per km travelled (d km). Write an expression for the total cost, and find the cost for a 12 km ride.

17. A number increased by 7 and then doubled gives 30. Write this as an equation and find the number.

18. Two numbers differ by 5, and the smaller number is n. Write an expression for the sum of the two numbers.

19. Apples cost Rs. a per kg, and you buy 3 kg. Write an expression for the total cost. If $a=120$, find the total cost.

20. Your monthly pocket money is Rs. m. If you save half of it every month, write an expression for your savings after 6 months.

Reflection Question: Why do you think mathematicians use letters to represent unknown numbers instead of always using specific numbers? Write your own idea.

Real-Life Application Question: Think of a real situation at home (cost of items, ages, distances) where a quantity can change. Write an algebraic expression to represent it using a letter.